

Week 11 — Achievers / HOTS Mixed Practice — Answer Key

Q1. C) 300 — If the original number is x : $1.15x = 345 \rightarrow x = 300$.

Q2. B) 21 — Four consecutive odd numbers $n, n+2, n+4, n+6$ sum to $4n+12 = 96 \rightarrow n = 21$.

Q3. B) ₹5,400 — Perimeter = $2 \times (36+24) = 120$ m; cost = $120 \times ₹45 = ₹5,400$.

Q4. B) ₹1,000 — Let the total be T . After books, $\frac{3}{5}T$ remains. After the bag: $\frac{3}{5}T - 240 = 360 \rightarrow \frac{3}{5}T = 600 \rightarrow T = 1,000$.

Q5. B) 5 years — Half the principal = ₹6,000. $SI = (P \times R \times T) \div 100 \rightarrow 6000 = (12000 \times 10 \times T) \div 100 = 1200T \rightarrow T = 5$.

Q6. C) 63 — Each term is "double, then add 1": $1 \times 2 + 1 = 3, 3 \times 2 + 1 = 7, 7 \times 2 + 1 = 15, 15 \times 2 + 1 = 31, 31 \times 2 + 1 = 63$.

Q7. C) 283 — Number = (divisor $\times$ quotient) + remainder = $(12 \times 23) + 7 = 276 + 7 = 283$.

Q8. C) 648 — $36 \times 18 = 648$.

Q9. C) ₹600 — Selling price = $CP \times 1.2 = 450 \times 1.2 = ₹540$. Since SP is 90% of the marked price (after a 10% discount): $MP \times 0.9 = 540 \rightarrow MP = 540 \div 0.9 = ₹600$.

Q10. C) ₹825 — Let the first friend get x ; the amounts are $x, x+150, x+300, x+450$, summing to $4x+900 = 2,400 \rightarrow x = 375$. The fourth friend gets $375+450 = ₹825$.

Q11. B) 12 cm — Square's perimeter = $4 \times 18 = 72$ cm. For the rectangle: $2 \times (24 + \text{breadth}) = 72 \rightarrow \text{breadth} = 36 - 24 = 12$ cm.

Q12. C) 24 — Total = people ahead + people behind + Rani herself = $14 + 9 + 1 = 24$.

Q13. C) 3,000 L — Combined rate = $450 + 300 = 750$ L/hour; in 4 hours = $750 \times 4 = 3,000$ L.

Q14. B) 12:17 — Smaller number = $1,450 - 850 = 600$. Ratio 600:850 — dividing both by their HCF (50) gives 12:17.

Q15. A) 74 — Let the number be $10a+b$ with $a > b, a-b=3, a+b=11$. Adding these: $2a=14 \rightarrow a=7$, so $b=4$. The number is 74. Check: $7-4=3 \checkmark, 7+4=11 \checkmark$.

Scoring: 12+/15 is a strong Achievers-section result — genuinely olympiad-level performance at this stage. Below that, go back to whichever week the missed questions came from before attempting Sample Paper 1.
